

Week 2 — Six Questions for Reading a Methods Section

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

Week 2

Six questions for reading a Methods section

Know what you are trying to rebuild, before you try to rebuild it.

- ▶ One argument, not two halves: the questions come first, NHANES arrives where a question cannot be answered without it
- ▶ Feeds **L1** (Thu) and **M0** shortlisting (due Wk3)

Why the question comes before the data

- ▶ The instinct is “what did they run?” — a model family answers nothing on its own
- ▶ A **Cox model appears under four different research goals**
- ▶ Reading a Methods section *is* the first act of reproduction

By the end of today: this paper answers four of six questions clearly, one partially, and one not at all.

The six questions

1. What is the **research goal**?
2. Who is the **population**, and how is eligibility operationalised?
3. What is the **exposure / index factor**, and the comparison?
4. What is the **outcome**, time zero, and follow-up?
5. What **other variables** enter, and what role are they supposed to play?
6. What **design and analysis features** must be respected?
If any one is under-specified, the study is not reconstructable — no matter how clean the results look.

Q1 — What is the research goal?

The paper states two aims, not one

- ▶ **Aim 1** — the *clinical correlates* of four groups defined by BMI and waist-to-height ratio
- ▶ **Aim 2** — the *prognostic value* of that classification for mortality

Related, but not the same question: the first is cross-sectional at baseline, the second needs follow-up.

Five goals the same variables can serve

Goal	What it asks
Description	How do the four groups differ, and what is their mortality experience?
Association	Is phenotype statistically associated with subsequent mortality?
Prognostic factor	Does phenotype inform mortality over and above other known factors?
Prediction	How accurately can we predict individual risk; does WHtR improve it?
Causal	What would mortality have been under a specified change in central adiposity?

Different goals need different evidence

- ▶ An **HR with a CI** is enough for an adjusted association or a prognostic effect
- ▶ It is **not** evidence a prediction model is good — that needs calibration and discrimination
- ▶ And not by itself evidence a causal effect was identified

The paper names “prognostic value” and supplies adjusted association.

Q2 — Who is the population?

NHANES in one picture

- ▶ Continuous NHANES in **2-year cycles**; ~**5,000 examined persons per cycle**; six cycles stacked here (2007–2018)
- ▶ Three collection settings — **Interview** (home) · **MEC exam** (measured body size, BP) · **Laboratory** (triglycerides, GGT, glucose)
- ▶ Data arrive as **many small topic files**, not one table

SEQN is the join key

- ▶ Every respondent has one **SEQN** — the identifier CDC uses to sort and merge files
- ▶ Analysis = pick the topic files you need, **merge on SEQN**, one row per person
- ▶ SEQN does **not** repeat across cycles: all **59,842** values across 2007–2018 are distinct and the per-cycle ranges do not overlap — *analysis set: all NHANES 2007–2018 records, $N = 59,842$*
- ▶ Keep a **cycle variable** anyway — provenance, changing definitions, per-cycle weights. Not to disambiguate SEQN

“MASLD” is a derived variable, not a diagnosis

- ▶ MASLD = hepatic steatosis **plus** 1 cardiometabolic criterion
- ▶ Gold-standard steatosis is biopsy or imaging — **NHANES has neither** in this window
- ▶ So: the **Fatty Liver Index**, thresholded at **FLI 60**, computed from **triglycerides, BMI, GGT and waist**

The disease definition forces the workaround. It is a feature of the data, not a shortcut.

The funnel, with a type on every row

Step	N	Kind of step
NHANES 2007–2018 records	59,842	—
Age ≥ 18	36,580	eligibility
In fasting subsample	14,962	design — drops 21,618
FLI computable	14,170	missingness
Steatosis (FLI ≥ 60)	6,052	target population
MASLD (≥ 1 criterion)	6,050	target population
Mortality-linked	6,048	outcome ascertainment

Analysis set: locked mortality-linked cohort, $N = 6,048$.

BREAK

Q3 — Exposure and comparison

The exposure is constructed, not measured

- ▶ Obesity by **BMI** ≥ 30 (Asian participants ≥ 25) $\times$ central adiposity by **WHtR** ≥ 0.60
- ▶ Four phenotype groups; **Group IV** = not obese by BMI, high waist-for-height
- ▶ The exposure spans **two collection settings** — BMI and waist both from the MEC exam, so a person missing either is missing the exposure

Which group is the comparison?

- ▶ No treated and control arm — the four-category phenotype is the **index factor**
- ▶ Mortality is compared **across** the categories
- ▶ **Group I is the reference.** Every hazard ratio is measured against it

A ratio without its referent means nothing — and here the referent is an obese group.

Eligibility and exposure share their inputs

Inputs	
Eligibility (FLI)	triglycerides, GGT, BMI , waist
Exposure (phenotype)	BMI , waist $\div$ height

*Being selected into the study and being placed in a phenotype group are **not independent events**. That is **structural dependence**.*

Never trust a label you have not recomputed.

Q4 — Outcome, time zero, follow-up

Outcome and the clock

- ▶ **All-cause mortality** (MORTSTAT) via the NCHS linked mortality file — *not* an NHANES file, a separate linked fixed-width one
- ▶ **Time zero is the MEC examination**; PERMTH_EXM is person-months from that exam
- ▶ Follow-up through **31 December 2019**, stated in the NCHS 2019 Linked Mortality File linkage methodology document — not inferred from the filename
- ▶ Entry spans 2007–2018, so follow-up lengths differ → survival analysis, not a risk difference

Time zero is a question the checklist never asks

- ▶ “Exposure, outcome, covariates” can all be specified while **time zero stays ambiguous**
- ▶ An ambiguous time zero silently changes who is at risk, and for how long
- ▶ Here exposure is measured **at** time zero — convenient, and rarer than you would think

Ask of your M0 candidates: when does follow-up begin, and does everyone start together?

Q5 — Other variables, and their roles

“In the model” is not the same as “confounding”

- ▶ The paper adjusts for age, sex, comorbidities, medications
- ▶ **“We adjusted for X” says where X sits in the syntax. It does not say what X does.**
- ▶ The same variable can be a confounder, a mediator, a competing prognostic factor, or a collider — and which it is depends on the **research goal from Q1**

What a crude-to-adjusted change establishes

- ▶ Group IV all-cause HR **15.1 crude** → **2.9 adjusted**
- ▶ Conditioning on the adjustment set moves the estimate **enormously**
- ▶ What that movement *means* — confounding controlled, mediation removed, a collider opened — depends on the goal and the causal role of those variables

Analysis set: full reconstructed analytic file, $N = 6,371$.

Q6 — Design and analysis features

NHANES is not a simple random sample

- ▶ Complex multistage probability design: **weights**, **strata** (SDMVSTRA), **PSUs** (SDMVPSU)
- ▶ We *use* them in **Week 6**; today, note they exist and that ignoring them is a choice
- ▶ Fasting-subsample analyses carry their own weight — WTSAF2YR, not WTMEC2YR

Designed subsampling is not missingness

- ▶ The fasting subsample is **statistically designed**, not self-selected, and carries a weight built to keep it nationally representative
- ▶ Check rather than assume: adults in and out of it differ by **0.2 years of age, 0.02 kg/m² BMI, 0.07 cm waist**
- ▶ It is balanced on the very measurements this paper's exposure is built from

Analysis set: all NHANES adults 2007–2018, $N = 36,580$.

Designed subsampling ordinary missingness.

Closing

M0 — ask the six of your own papers

- ▶ Shortlist two or three candidate NHANES papers; answer **all six**, each with a pointer to where in the paper you found it
- ▶ **Reconstructable is necessary, not sufficient** — a perfectly specified paper can still be a poor M0 choice
- ▶ M0 wants: enough observations and events, a design feature its analysis must respect, and a missing variable the analysis actually uses
- ▶ An unanswered question is not a disqualification — it is a candidate for the judgement you defend at **M1**

Bridge to the lab and M0

- ▶ **In L1 (Thu):** download several cycles, read the raw topic files, merge on SEQN into one analytic-ready table with a cycle indicator
- ▶ **How it becomes your project:** L1's merge is the substrate for the whole term
- ▶ **Toward M0 (Wk3):** bring six answers per candidate paper, not a summary

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